

Santiago Hopkins

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EDUCATION

University of California, Berkeley

B.S. Mechanical Engineering, Minor in Computer Science

NCAA Division 1 Student-Athlete (Men's Soccer)

GPA: 3.82/4.0

EXPERIENCES

Triton Robotics - Robotics Systems Engineering Intern

01/2025 - Present

2-Axis Robot for Composite Tube Fabrication

- Led fabrication automation of 1.5 meter composite outer tube for dynamic rigidization via 2-axis robot, decreased fabrication time by 70%
- Integrated rotary stage to allow for $\pm 0.5^\circ$ adjustment of filament fleet angle, integrated analog magnetic brakes for tape-backing removal
- Selected industrial, fully-integrated stepper motors for both spindle and linear axes, controlling for speed and torque, respectively, via PID
- Performed 5 time studies, selecting greatest time sinks to influence hot-fixes in preparation for FIH trials, and larger-scope concept changes

Zoox - Mechatronics Engineering Intern

05/2024 - 08/2024

Hydro-Pneumatic Sensor-Pod Wash System

- Led redesign of hydro pneumatic circuit, developed Python script to report the percentage of area cleaned, achieved 98% cleaning efficiency
- Developed injection molded mounts and interfaces for alternate supplier external gear water pumps and normally-closed solenoid valves
- Integrated CAN communication for solenoid valve control and LIN comm. for pump control, enabling 10hz pump motor state messaging
- Designed a 1-D hydraulic circuit simulation in GT-Suite, incorporating localized & distributed losses, achieved approximately 88% accuracy

Water Contact Angle Measurement Software Automation Tool

- Developed design criteria and core functionality for an automated WCA measurement tool to accurately assess solar and natural degradation of hydrophobic coatings on onboard camera lenses, both in controlled lab environments and during routine maintenance checks
- Implemented using Python and C++, leveraging OpenCV2 for computer vision tasks and the LSQ-Ellipse library for precise ellipse fitting
- Applied Gage R&R methods to validate on-fleet WCA automation tool, achieving $\pm 5^\circ$ accuracy, and reducing processing time by $\sim 60\%$

Formula Electric at Berkeley (FSAE Driverless) – Autonomous Robotic Mechanisms Lead

01/2023 - 09/2024

- Selected and spearheaded a team of 4 in the first iteration of the robotic systems that would steer and brake a Formula Student Style EV in an attempt to be the first ever American student team to have a fully-autonomous race car; originally, a part of FSAE Germany competition

Autonomous Steering System

- Led development and integration of modular robotic steering system to mimic human steering inputs, achieving lock to lock steering in $< 1s$
- Specified system criteria, targeting 9Nm of torque and $> 100RPM$ for motor selection, retaining low-profile design and minimizing backlash
- Selected RMDx8 servo with high coil count, 1:6 planetary gearbox, and dual encoder. Verified output, finding optimal operating voltage
- Designed belt drive system with custom HTD-5mm pulleys, and built an in-house hall-effect absolute rotary encoder to prevent overturning
- Integrated driving motor via CAN communication, and configured microcontroller to perform rapid adjustments via a tuned PID controller

Autonomous Braking System

- Led development and integration of modular pneumatic braking system, achieving 100 actuations w/ independent control of front, rear brakes
- Designed an Emergency Brake System (EBS) via mechanical and electrical methods, enhancing existing fail safes for a 180ms response time
- Developed pneumatic circuit using pistons @ 7 Bar to pressurize front, rear brake hose in parallel w/ brake pedal, retaining manual operation
- Developed robust, low-level PID controller (C++) for variable pressure regulator, minimizing deceleration error and integral wind-up

World Surf League | Kelly Slater Wave Co. – Wave Systems Test Engineering Intern

05/2023 -12/2023

CFD Prototyping

- Led CAD development and on-site implementation of experimental methods to validate CFD metrics, such as resultant force, vortex ratio
- Analyzed wave video to determine the effect of on-pole LED strip lights on barrel-vortex ratio (BVR), reported $\sim 3\%$ BVR error

PROJECTS

DIY Autonomous Drone

07/2024 - 12/2024

- Developed path & motion planning algorithms within ROS2 nodes, achieved controlled motion over a 1m distance at a height of 25cm
- Implemented rudimentary nearest neighbors algorithm for global path planning, implemented 'carrot-on-a-stick' method for motion planning
- Developed 3D printed fixtures to protect Raspberry Pi and Arduino Microcontroller, maintaining symmetry and minimizing weight
- Developed robust test fixtures to constrain drone in one axis to streamline Nonlinear PID tuning, programmed fail safes and debugging node

Berkeley Institute of Transportation Studies | Smart Cities – Undergraduate Research Assistant

05/2023 - 08/2023

- Using VTA traffic data, developed a robust Python model visualized with Kepler.gl to analyze traffic patterns in a scenario w/o public transit
- Began initial data acquisition and modeling for a scenario where the City of Berkeley removed 'Berkeley Barriers' from certain intersections

SKILLS

Hardware: SolidWorks, CATIA v5/v6, Siemens NX, Onshape, Ansys, GT-Suite, Star-CCM+, DFM/A, Altium

Software: Python, C++, ROS2, Git, MATLAB, Node JS, C, Java

Manufacturing: GD&T, CNC Machining, Injection Molding, Casting, Waterjet, Laser Cutter, FDM/SLA 3D Printing, Lathe, Mill, Drill Press